Sep 22, 2026 (Due: 08:00 Sep 29, 2026)

1. Assume that b, Δb, x, Δx satisfy

$$Ax = b, \qquad A(x + \Delta x) = b + \Delta b,$$

where

$$A = \begin{bmatrix} 987 & 1597 \\ 1597 & 2584 \end{bmatrix}.$$

Construct examples such that
(1) $\|\Delta b\|_\infty / \|b\|_\infty$ is very small while $\|\Delta x\|_\infty / \|x\|_\infty$ is very large;
(2) $\|\Delta b\|_\infty / \|b\|_\infty$ is very large while $\|\Delta x\|_\infty / \|x\|_\infty$ is very small.

2. Let $\hat{x}$ be an approximate solution of the nonsingular linear system $Ax = b$. The vector $r = b - A\hat{x}$, known as the *residual*, can be treated as a backward error on b because $A\hat{x} = b - r$. Try to construct a matrix E such that $(A + E)\hat{x} = b$ and

$$\frac{\|E\|}{\|A\|} = O\Big(\frac{\|r\|}{\|b\|}\Big).$$

3. It can be shown that Gaussian elimination with partial pivoting is numerically stable for solving nonsingular tridiagonal linear systems, in the sense that the growth factor is bounded. Give a concrete upper bound on the growth factor.

Note: Let us use the classical definition of the growth factor:

$$\rho = \max_{i,j,k} \frac{a_{i,j}^{(k)}}{\|A\|_\infty},$$

where $a_{i,j}^{(k)}$ represents the matrix element $a_{i,j}$ after k steps of Gaussian elimination.

4. We have mentioned in the lecture that for complex vectors x, $y \in \mathbb{C}^n$, $\|x\|_2 = \|y\|_2 > 0$ does *not* guarantee that there exists a Householder reflection H such that $y = Hx$. Provide a sufficient and necessary condition for the existence of a Householder reflection H so that $y = Hx$. Prove your claim.

5. Solve the following linear systems, using Gaussian elimination with and without pivoting:

$$\begin{bmatrix} 8 & 1 & & & & & \\ 6 & 8 & 1 & & & & \\ & 6 & 8 & 1 & & & \\ & & \ddots & \ddots & \ddots & & \\ & & & 6 & 8 & 1 & \\ & & & & 6 & 8 & 1 \\ & & & & & 6 & 8 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_{98} \\ x_{99} \\ x_{100} \end{bmatrix} = \begin{bmatrix} 9 \\ 15 \\ 15 \\ \vdots \\ 15 \\ 15 \\ 14 \end{bmatrix}$$

and

$$\begin{bmatrix} 6 & 1 & & & & & \\ 8 & 6 & 1 & & & & \\ & 8 & 6 & 1 & & & \\ & & \ddots & \ddots & \ddots & & \\ & & & 8 & 6 & 1 & \\ & & & & 8 & 6 & 1 \\ & & & & & 8 & 6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_{98} \\ x_{99} \\ x_{100} \end{bmatrix} = \begin{bmatrix} 7 \\ 15 \\ 15 \\ \vdots \\ 15 \\ 15 \\ 14 \end{bmatrix}.$$

Compare the computed solutions with the exact ones. How can you say about the accuracy?

6. (optional) It can be shown that Gaussian elimination without pivoting is numerically stable for solving strictly diagonally dominant linear systems, in the sense that the growth factor is bounded. Give a concrete upper bound on the growth factor.

7. (optional) The $n \times n$ Hilbert matrix H_n is ill-conditioned if n is not too small. Calculate or estimate the condition number of H_n.